

LLD-01 – RAN Equipment Design (vendor-neutral)

Low-level design of radio, baseband, antenna-line and site power equipment for all 6,270 sites. Per-site result: `lld/ran_site_equipment.csv` ; per-hub baseband pools: `lld/ran_hub_baseband.csv` . Unit capacities and power figures are typical 2026 carrier-grade values – replace with the chosen vendor's datasheet in `tools/lld_design.py` and re-run.

1. Equipment standards

Code	Equipment	Key specification	Avg / peak power	Network quantity
AAU-64	n78 64T64R massive-MIMO AAU	3400–3800 MHz, 400 MHz IBW / 300 MHz OBW, 320 W, 192 elements, 24 dBi, eCPRI 3 × 25GE, ≤ 30 kg, ≤ 0.5 m ² , IP65, –40...+55 °C	900 / 1,300 W	17,076
R-8T	n78 8T8R radio + 8-port antenna (rural)	8 × 40 W, 300 MHz OBW, 17.5 dBi	750 / 1,050 W	1,331
RRU-L2	n28 2T4R RRU (city)	2 × 60 W, 703–748 / 758–803 MHz	350 / 500 W	17,076
RRU-L4	n28 4T4R RRU (rural / highway)	4 × 40 W	550 / 750 W	1,599
RRU-M	n1 + n3 dual-band 4T4R RRU	4 × 60 W shared, 1800 + 2100 MHz	700 / 950 W	11,672
RRU-M3	n3 4T4R RRU (suburban / highway)	4 × 40 W	450 / 620 W	7,003
ANT	Passive antenna 2L4H + RET	2 × 698–960 (15.5 dBi) + 4 × 1695–2690 MHz (17.5 dBi), 2.0 m city / 2.6 m rural, AISG 2.0 RET per band	–	18,675
DU	Distributed unit (baseband)	≥ 9 × 100 MHz 64T64R carriers + 12 FDD cells, 6 × 25GE fronthaul, 2 × 25GE midhaul, Class-C PTP, GNSS	350 / 500 W	2,371 at site + 2,960 pooled
FHGW	Fronthaul gateway / passive WDM mux (C-RAN site)	12-λ MWDM/LWDM 25G, CPRI→eCPRI conversion for FDD RRUs, site alarms	60 / 80 W	3,899
CSR	Cell-site router (D-RAN site)	4 × 25GE + 8 × 10GE, SR-MPLS/SRv6, Class-C boundary clock, SyncE	120 / 150 W	2,371

Total NR cells: **104,243** (city sector = 3 × n78 + n28 + n1 + n3; suburban without n1; rural = 3 × n78 + n28 + n1 + n3; highway = n28 + n3). Software licences follow cells, carriers, MIMO layers and CA combinations (n3 PCell + n1 + 3 × n78 DL CA; UL on n78 or FDD).

2. Standard site configurations

Configuration	Sites	Cells	Avg load W	Peak load W	Rectifier kW (N+1)	Li-ion battery kWh	Grid kVA
U C-RAN 3-sector	2782	18	6,060	8,530	21	33.6	15
SU D-RAN 3-sector	1605	15	5,720	8,110	21	33.6	15
SU C-RAN 3-sector	640	15	5,310	7,540	18	28.8	15
DU C-RAN 3-sector	477	18	6,060	8,530	21	33.6	15
R D-RAN 3-sector	443	18	6,620	9,100	30	72.0	15
U D-RAN 3-sector	188	18	6,470	9,100	21	38.4	15
HW D-RAN 2-sector	134	4	2,620	3,590	15	28.8	5

- DC power –48 V, 3 kW rectifier modules, N+1, sized for peak load + 0.2C battery recharge. Battery autonomy **4 h city, 8 h rural / highway** at average load (80 % depth of discharge). Li-ion with anti-theft lock and remote BMS.
- **360 rural / highway sites without fibre or grid nearby are designed off-grid:** solar PV ≈ 23.1 kWp (traffic-adaptive radio sleep, 5.5 peak-sun-hours) + battery + 15 kVA stand-by generator.
- Energy saving features mandatory in the tender: symbol / channel shutdown, carrier shutdown of 2nd–3rd n78 carrier at night, deep-sleep AAU – target ≥ 25 % energy reduction versus the averages above.
- Outdoor cabinets IP55 with heat exchanger (no air-conditioning) up to 50 °C ambient; Aqaba and Jordan Valley sites use the 55 °C variant with sun shield.

3. Baseband architecture

- **Dense urban / urban: C-RAN.** AAUs and RRUs connect over passive-WDM fronthaul (≤ 10 km, ≤ 75 μ s one-way, 12-core drop) to the DU pool in their hub. Pool size = 0.75 DU per site (25 % pooling gain from non-coincident busy hours); inter-site CA, DL CoMP and coordinated scheduling run inside the pool – this is what lifts the cell-edge pixels to 1 Gbps.
- **Suburban / rural / highway: D-RAN.** DU in the site cabinet, 25GE midhaul on the access ring.
- **CU (CU-CP / CU-UP) virtualised** on the edge cloud at the six core / edge sites; one CU cluster per ≈ 500 sites; F1 over the aggregation network, latency budget ≤ 5 ms.
- **150 hubs**, of which 93 are DU hotels with **2,960 pooled DUs**; largest hub 54 DUs / 7 racks / 43.7 kW facility load. Hub rooms: 2N power feeds, N+1 cooling, 4 h battery + generator, GNSS antenna pair, fire suppression, access control.

4. Antenna line and installation standards

Jumpers ≤ 3 m, 1/2" super-flex, 4.3-10 connectors, PIM ≤ -153 dBc; AISG RET daisy-chain per sector; AAU on independent bracket with $\pm 15^\circ$ mechanical azimuth and 0–10° tilt adjustment; wind load design 160 km/h; rooftop pole loading checked for 75 kg per sector (AAU + passive antenna + RRUs). GNSS receiver per D-RAN site and per hub; PTP as backup (and primary for C-RAN radios).

5. Parameter baseline (common to all vendors)

SCS 30 kHz (n78) / 15 kHz (FDD); TDD pattern DDDSU, special slot 10:2:2; SSB 8 beams n78; PCI from `output/sectors.csv` ; digital tilt and RET per sector from the same file; max 4 DL layers per UE, MU-MIMO up to 16 layers; 256-QAM DL / 64-QAM UL (256-QAM UL where supported); n3 PCell with n78 SCells, UL Tx switching enabled; VoNR on n28/n3 with EVS-WB; inactivity timer 10 s; A3 offset 3 dB, TTT 320 ms (city) / 640 ms (rural).

LLD-02 – 5G Core Network Design (vendor-neutral)

Cloud-native 5GC (3GPP Rel-17 SBA) sized for **4,204,435 subscribers** and **2,492 Gbps busy-hour user plane**.

Tables: `lld/core_nf_sizing.csv` , `lld/core_site_sizing.csv` .

1. Dimensioning inputs

Parameter	Value
Provisioned subscribers	4,204,435
Simultaneously registered	85 % → 3,573,770
PDU sessions per registered sub	1.6 (internet + IMS) → 5,718,032
Busy-hour user plane	2,492 Gbps average, × 1.2 peak-to-mean
VoNR subscribers	60 % → 63,067 Erlang (1 BHCA, 90 s)
Design utilisation	70 % of rated capacity, N+1 per function, geo-redundant 1+1 between DC1 and DC2

Busy-hour signalling load (transactions per second): registration 1,191, pdu session 2,978, service request 11,913, handover 7,942, paging 3,971 → **27,995 procedures/s**, ≈ 167,967 SBI transactions/s.

2. Network-function sizing (per data centre – each DC carries 100 % on failure of the other)

Network function	Driver	Unit	Capacity / instance	Instances (N+1)	vCPU
AMF	3,573,770	registered subs	2,000,000	4	2560
SMF	5,718,032	PDU sessions	3,000,000	4	2880
AUSF + UDM	4,204,435	provisioned subs	5,000,000	3	1440
UDR (subscriber DB)	4,204,435	provisioned subs, 3 replicas	10,000,000	2	1920
PCF + BSF	5,718,032	policy sessions	3,000,000	4	2240
CHF (charging)	5,718,032	charging sessions	4,000,000	4	2560
NRF + NSSF + SCP	167,967	SBI transactions/s (≈ 6 per procedure)	60,000	5	1600
NEF + NWDAF	4,204,435	exposure / analytics	10,000,000	2	960
IMS (P/I/S-CSCF, TAS, MRF, SBC)	2,522,661	VoNR subs	1,500,000	4	4400
SEPP + roaming / interconnect GW	210,222	roamers	1,000,000	2	480
SMSF + CBCF (SMS, public warning)	4,204,435	subs	10,000,000	2	400
OSS, observability, LI mediation, security	4,204,435	platform	5,000,000	3	2700

Control plane per DC: **24,140 vCPU** + 25 % platform overhead → **315 servers** (2-socket, 96 usable vCPU, 512 GB RAM, 2 × 100GE). Subscriber database (UDR) synchronously replicated DC1↔DC2, asynchronous third replica in Irbid.

3. Site sizing (user plane distributed, control plane central)

Site	UP share	Peak Gbps	UPF servers (N+1)	CP servers	Total servers	Racks	IT load kW	400GE ports to transport
DC1 Amman West	27.5 %	822	8	315	323	23	254	6

Site	UP share	Peak Gbps	UPF servers (N+1)	CP servers	Total servers	Racks	IT load kW	400GE ports to transport
DC2 Amman East	27.5 %	822	8	315	323	23	254	6
Amman Centre edge	8.0 %	239	3	0	9	3	19	2
Zarqa edge	7.0 %	209	3	0	9	3	19	2
Irbid regional DC	15.0 %	449	5	0	17	4	25	4
Aqaba regional DC	8.0 %	239	3	0	15	3	23	2
Regional PoPs (Karak, Ma'an, Mafrq)	7.0 %	209	3	0	9	3	19	2

UPF server: 2 × 100GE SmartNIC, ≈ 180 Gbps per server at 70 % load with full feature set (charging, LI, DPI-lite).
UPF selection by TAC / DNAI so traffic breaks out at the nearest site; CDN caches and MEC hosts co-located on the N6 side of every UPF site.

4. Logical design

- **PLMN** 416-xx (MCC 416 Jordan; MNC from TRC). **TAC**: one per hub cluster (150 + rural TACs), TA lists sized ≤ 16 TACs to keep paging below 3,971 /s.
- **Slices (S-NSSAI)**: SST 1 / SD 000001 eMBB-1G (GfBR 1 Gbps tariff, 5QI 6/8/9 + dedicated GBR flow for the premium tier) · SST 1 / SD 000002 FWA · SST 1 / SD 000010 enterprise / private DNNs · SST 2 URLLC (future) · SST 1 / SD 0000FF public safety · O&M.
- **DNNs**: internet, ims, fwa, enterprise-, iot, sos.
- **Addressing**: IPv6-first. UE pools: one /40 per UPF site from the operator's RIPE NCC /29–/32 allocation (a /64 per UE session), IPv4 via NAT64/464XLAT and a CGNAT pool (≈ 1 public IPv4 per 64 users).
Infrastructure: loopbacks 10.255.0.0/16, SBI 10.10.0.0/16 per DC, N2/N3 10.20.0.0/14 regionalised, O&M 10.250.0.0/16, all dual-stack with IPv6 ULA-free GUA on N3.
- **Naming / discovery**: 3GPP FQDNs (`amf1.set01.region01.amf.5gc.mnc0xx.mcc416.3gppnetwork.org`), NRF hierarchical (one per DC + PLMN-level), SCP model C for indirect communication, DNS anycast pairs in every site.
- **Voice**: IMS with VoNR only (no LTE layer in a pure-SA network) – EVS-WB, SRVCC not applicable; emergency calls on n28/n3; interconnect to other operators through SBC pair in DC1/DC2 (SIP-I / SIP), roaming through SEPP (N32) and IPX; devices without VoNR are a launch risk (see deployment risk register).
- **Security**: zones Untrust (N6/internet) – DMZ (SEPP, SBC, NEF) – Core SBA – O&M – LI; mutual TLS on all SBI with operator PKI; SUCI concealment (profile A) enabled; N2/N3 protected by IPsec from D-RAN sites / MACsec on owned fibre; firewalls + DDoS scrubbing on N6 (≥ 20 % of N6 capacity); LI (X1/X2/X3) mediation in both DCs.
- **Cloud platform**: Kubernetes CaaS on bare metal, SR-IOV / DPDK for UPF, separate clusters per zone, GitOps lifecycle, in-service upgrade with canary; observability (metrics, traces, logs) and NWDAF feeding the capacity trigger (sector PRB > 20 %).
- **Charging / policy**: converged CHF (online + offline), PCF rules for fair-use shaping that protects the single-user 1 Gbps experience; BSS integration over REST/TMF APIs.

5. Build phases

C0 facilities (M1–M2) → C1 control plane, UPF pool 1, IMS, first call (M2–M4) → C2 geo-redundancy tests, Amman Centre + Zarqa edge (M6–M9) → C3 Irbid and Aqaba regional DCs (M8–M12) → C4 UPF capacity steps of ≈ 400 Gbps per 600 accepted city sites (M13–M30) → C5 final dimensioning, DR drill, hand-over (M31–M36).

LLD-03 – Transport Network Design (vendor-neutral)

Ports and router classes per node, optics, DWDM wavelength plan, IP/MPLS design and microwave link design.

Tables: `transport_hub_ports.csv`, `transport_pop_ports.csv`, `transport_agg_ring_optics.csv`, `dwdm_wavelength_plan.csv`, `microwave_links.csv`.

1. Node classes

Class	Where	Ports	Quantity
CSR	every D-RAN site	4 × 25GE + 8 × 10GE, Class-C BC	2,371
FHGW / passive WDM	every C-RAN site	12 λ × 25G on one fibre pair (+ protection pair)	3,899
HUB-R1	hub (pair per hub)	24×25GE + 4×100GE	117 hubs → 234 routers
HUB-R2	hub (pair per hub)	48×25GE + 8×100GE	33 hubs → 66 routers
POP-R1	PoP / core site (pair)	16×100GE + 4×400GE	24 PoPs → 48 routers
DWDM	backbone nodes	ROADM / FOADM, 400G coherent line cards	15 routes, 37 wavelengths day 1 (400G on rings, 100G on light spurs)

2. Access and aggregation dimensioning

- **Access ring (D-RAN), 25GE:** ≤ 8 sites, dual-homed on the hub router pair. Design load = 1.5 × busy-hour mean + one sector at its 5.2 Gbps peak. Result over 269 D-RAN rings: mean 7.9 Gbps, max 10.3 Gbps → ≤ **41 % of 25GE**; upgrade path 50GE optics on the same fibre.
- **C-RAN access ring:** physical fibre ring carrying point-to-point fronthaul wavelengths (9 × 25G eCPRI for the AAUs + 3 × 10G for FDD RRUs per site, passive WDM, both directions of the ring for protection). Optical budget: 25G LR ≤ 10 km, ≤ 6.3 dB path + 2 × 2.5 dB mux.
- **Hub routers:** one end of every access ring per router + half of the pooled DUs + 100GE uplinks; peak hub load max 70.5 Gbps, uplink utilisation at peak ≤ 29 %.
- **Aggregation rings, 100GE:** ≤ 5 hubs per ring; optic by mean span: 39 × 100G-LR4 (10 km), 2 × 100G-ER4 (40 km). Ring load = sum of hub peaks × 0.7 diversity; rings above 60 Gbps are split or moved to 2 × 100GE LAG (check column peak_Gbps per PoP).
- **PoP routers:** aggregation rings on 100GE, 400GE to DWDM / core; busiest PoP POP-IRB 298.1 Gbps busy hour.

3. DWDM backbone – wavelength plan (day 1, 400G coherent, 75 GHz flex-grid, 64 slots per fibre pair)

Route	Ring	km	Design demand Gbps	λ day 1	Channel plan	In-line amplifier sites	Node type	Line optics
Desert Highway (R15) Amman-Aqaba	Southern ring	340	1340	4	4 × 400G (of 64 × 75 GHz C-band slots)	4	ROADM (2-degree, 3 at junctions)	400G ZR+/OTN coherent, FlexO

Route	Ring	km	Design demand Gbps	λ day 1	Channel plan	In-line amplifier sites	Node type	Line optics
Dead Sea - Wadi Araba (R65) Amman-Aqaba	Southern ring	337	1340	4	4 × 400G (of 64 × 75 GHz C-band slots)	4	ROADM (2-degree, 3 at junctions)	400G ZR+/OTN coherent, FlexO
Amman-Zarqa-Mafraq-Jaber	Northern ring	74	1039	3	3 × 400G (of 64 × 75 GHz C-band slots)	0	ROADM (2-degree, 3 at junctions)	400G ZR+/OTN coherent, FlexO
Mafraq-Ramtha-Irbid	Northern ring	52	1039	3	3 × 400G (of 64 × 75 GHz C-band slots)	0	ROADM (2-degree, 3 at junctions)	400G ZR+/OTN coherent, FlexO
Irbid-Jerash-Amman (R35)	Northern ring	68	1039	3	3 × 400G (of 64 × 75 GHz C-band slots)	0	ROADM (2-degree, 3 at junctions)	400G ZR+/OTN coherent, FlexO
King's Highway Madaba-Karak-Tafilah-Petra-Ma'an	spur / regional	241	509	2	2 × 400G (of 64 × 75 GHz C-band slots)	3	FOADM / terminal	400G ZR+/OTN coherent, FlexO
Jordan Valley (R65 north)	spur / regional	127	10	2	2 × 100G (of 64 × 75 GHz C-band slots)	1	FOADM / terminal	100G coherent
Amman-Salt-Deir Alla	spur / regional	37	145	2	2 × 100G (of 64 × 75 GHz C-band slots)	0	FOADM / terminal	100G coherent
Irbid-Ajloun-Jerash	spur / regional	47	79	2	2 × 100G (of 64 × 75 GHz C-band slots)	0	FOADM / terminal	100G coherent
Karak-Qatraneh	spur / regional	39	0	2	2 × 100G (of 64 × 75 GHz C-band slots)	0	FOADM / terminal	100G coherent
Tafilah-Jurf	spur / regional	34	0	2	2 × 100G (of 64 × 75 GHz C-band slots)	0	FOADM / terminal	100G coherent
Amman-Azraq (R40)	spur / regional	112	16	2	2 × 100G (of 64 × 75 GHz C-band slots)	1	FOADM / terminal	100G coherent
Zarqa-Azraq-Safawi	spur / regional	137	0	2	2 × 100G (of 64 × 75 GHz C-band slots)	1	FOADM / terminal	100G coherent
Mafraq-Safawi-Ruwaished (R10)	spur / regional	302	0	2	2 × 100G (of 64 × 75 GHz C-band slots)	3	FOADM / terminal	100G coherent
Quweira-Wadi Rum	spur / regional	32	0	2	2 × 100G (of 64 × 75 GHz C-band slots)	0	FOADM / terminal	100G coherent

Demand = busy-hour traffic of the clusters homed on the route × 1.3 peak × 1.5 growth; ring routes carry the **whole ring demand** so any single fibre cut is survivable (OTN 1+1 / ROADM restoration < 50 ms for protected services). The southern ring also carries ≈ 399 Gbps of international transit from the Aqaba cable landing (40 % of internet volume, after 60 % in-country cache hit). EDFA every ≤ 80 km (Raman on > 100 km desert spans), G.652.D fibre, span loss budget 0.22 dB/km + 0.5 dB per splice-km allowance; OSNR ≥ 24 dB end-to-end for 400G 16-QAM. Fill after 3 years ≈ 4 % of slots – ample headroom.

4. IP / MPLS design

- **IGP:** IS-IS L2, one instance per domain (metro Amman, north, south, east), multi-instance at PoP border routers; **SR-MPLS with TI-LFA (< 50 ms)**, SRv6-ready hardware; BFD 3 × 10 ms on ring links.
- **Services:** BGP EVPN / L3VPN per function – RAN-CP (N2), RAN-UP (N3), F1 (C-RAN midhaul), O&M, Sync, Enterprise; route reflectors in DC1 / DC2 / Irbid; inter-domain with BGP-LU + seamless MPLS.
- **Slicing:** FlexE / flex-algo hard slices for eMBB-1G (low-delay flex-algo), FWA, enterprise, O&M.
- **QoS (8 classes):** PTP/sync EF-strict · network control CS6 · RAN control + VoNR signalling CS5 · VoNR media EF · eMBB-1G GFBR AF41 · FWA / premium data AF31 · best-effort data BE · O&M AF21; 5QI→DSCP mapping done in DU/CU and UPF; ring links shaped at 95 %.
- **MTU 9,100** end-to-end (GTP-U + SRv6 headroom); IPv6 on all infrastructure links, IPv4 only where a legacy management system needs it.
- **Addressing:** loopbacks 10.255..0/24, p2p /31 from 10.254.0.0/16, IPv6 /127 from the infrastructure /40; site subnets summarised per hub.
- **Synchronisation:** ePRTC + GNSS at DC1, DC2, Irbid, Aqaba; PRTC-B at every hub; G.8275.1 full on-path support, every router Class-C boundary clock; budget ±1.5 µs network, ±130 ns relative inside a C-RAN cluster; SyncE everywhere; GNSS holdover 24 h at hubs.
- **Security:** MACsec on owned fibre rings, IPsec (N2/N3) from sites on leased or microwave links, control-plane policing, RTBH / flowspec, out-of-band management network via LTE-free DCN on dedicated VPN.

5. Microwave design (360 rural / highway sites)

Radio configuration	Hops
E-band + 18 GHz multiband (carrier-bonded)	218
18 GHz XPIC 2+0, 2 × 112 MHz, 2048-QAM	135
E-band 80 GHz 1+0, 2 GHz channel	7

- Per-hop budget in `microwave_links.csv` : free-space loss, antenna gain, receive level, fade margin, ITU-R P.530/P.838 rain attenuation at R0.01 = 22 mm/h, and a **terrain line-of-sight check on the Copernicus DEM** (60 % of the first Fresnel zone, $k = 4/3$, antennas 3 m below tower top).
- Result: **251 hops clear, 109 blocked by terrain** – those need a taller structure, a relay / passive repeater, or a different far end (column `terrain_line_of_sight` gives the missing clearance); 0 hops are rain-limited and take the next dish size.
- Hops longer than 16 km deliver ≤ 1.3 Gbps: those sites start with one 100 MHz n78 carrier on the rural 8T8R radio and move to fibre or a second hop when traffic requires.
- Frequency coordination and licensing with TRC per hop; E-band under light licensing where available; XPIC with ACM down to QPSK for 99.999 % of the control / voice traffic class.